

# UAV-Assisted Multi-Access Computation Offloading via Hybrid NOMA and FDMA in Marine Networks

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**Abstract**—With the rapid development of marine networks, there have been growing demands for computation-intensive and delay-sensitive marine applications and services. However, the limited underwater energy supply and the acoustic channels result in the low efficiency for computing tasks and high transmission delay. In this paper, we investigate the unmanned aerial vehicles (UAVs)-assisted multi-access computation offloading in marine networks, with the objective of minimizing the energy consumption of ocean devices. Specifically, for the underwater segment, we consider the scenario that multiple underwater sensor nodes (USNs) covered by the unmanned surface vehicle (USV) upload their sensing data via non-orthogonal multiple access (NOMA) for improving the channel utilization. For the radio frequency segment, we consider the scenario that multiple UAVs hovering in the air act as the aerial edge servers for providing computing services, in which the USV offloads the workloads to UAVs via frequency division multiple access (FDMA) for avoiding their co-channel interference, while taking into account that a malicious node overhears the USV's offloading transmission. To improve the computation offloading efficiency, we formulate an optimization problem for USNs and USV to minimize the total energy consumption by jointly optimizing the USN's uploading time, USV's computation offloading, USV's offloading time, and the secrecy provisioning. Despite the non-convexity of the formulated

joint optimization problem, we exploit a layered structure to decompose the problem, and propose efficient algorithms to obtain the optimal solutions. Finally, we conduct simulations to validate the effectiveness and efficiency of the proposed algorithms. Numerical results demonstrate that our algorithms can significantly reduce the energy consumption in comparison with the benchmark schemes.

**Index Terms**—Energy efficiency, marine networks, multi-access computation offloading.

## I. INTRODUCTION

WITH the growing demands for marine activities, there has been an increasing number of marine devices deployed in the ocean [1], [2], which leads to the requirements for high-speed and low-latency marine communications [3], [4]. For instance, the marine navigation data are required for safe navigation of vessels [5]. The real-time data collection is needed for safe operation of offshore drilling platforms [6]. The underwater sensor nodes (USNs) and buoys have a large amount of hydrological data to process. The real-time voices and videos are often required for maritime rescue. To this end, the computation offloading in marine networks is of great significance to meet the growing service requirements for marine activities, and have been paid much attention by researchers in recent years.

Thanks to the advent communication and computation techniques [7], marine edge computing, which enables oceanic data processing as close to the source as possible, has been envisioned as the promising paradigm to support energy efficiency and low latency communications [8], [9]. Specifically, the marine edge computing architecture can be divided into two transmission segments, i.e., the underwater transmission segment and the radio frequency transmission segment. For the underwater segment, it covers the communication and data transmission between USNs and surface nodes, and the acoustic transmission has been regarded as an effective approach for improving the data transmission efficiency [10]. For the radio frequency segment, it provides the communication and data transmission between surface nodes and terrestrial nodes for computation offloading. However, the underwater acoustic transmission suffers from low data rate, long propagation delay, and high packet loss probability, which reduces the on-

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demands services in marine applications [11]. In addition, the USNs are powered by batteries, which might be quickly exhausted due to the significant consumption for overcoming the acoustic signal attenuation. Moreover, USNs are hard to be recharged due to the complicated marine environment. Therefore, it is necessary to design an energy efficient computation offloading scheme to prolong the lifetime of USNs and guarantee the reliable data transmission.

Multi-access computation offloading in ocean environment is an effective approach to improve the computing efficiency of ocean devices [12], [13], [14]. Existing works focusing on oceanic computation offloading can be classified into two categories. One approach is to directly offload the workloads to unmanned devices (e.g., unmanned surface vehicles (USVs), autonomous underwater vehicles (AUVs)) [15]. This approach aims to save the energy consumption of USNs by path planning to shorten the data transmission distance. Another approach is to exploit the relay transmission to offload the workloads to terrestrial nodes (e.g., offshore base stations, unmanned aerial vehicles (UAVs)) [16] for processing through multi-hop routing schemes, which requires at least one relay node between USN and terrestrial nodes [17]. This approach mainly focuses on the relay selection and routing design to improve the relay performance. Different from these offloading modes, multi-access computation offloading enables ocean devices to offload their workloads to multiple edge nodes in marine networks, which has been expected to effectively enhance the computing efficiency.

Existing works adopt the time division multiple access (TDMA) and frequency division multiple access (FDMA) for data transmission between USNs and surface nodes in the underwater environment, since there is no co-channel interference among different acoustic channels [18]. However, the TDMA requires the strict time sequence for data transmission, which results in high cost on information exchanges. The acoustic bandwidth resource utilization in FDMA is low, which degrades the transmission throughput [19]. The non-orthogonal multiple access (NOMA) [20], [21], which allows USNs to reuse the same resource block for data transmission, has been considered as an efficient approach to improve the channel efficiency and transmission throughput. In addition, due to the nature of radio broadcasting in radio frequency segment, malicious nodes can intentionally eavesdrop the radio signals and decode the transmission data. Therefore, the state-of-art solutions for improving the data transmission efficiency and guaranteeing secrecy provisioning in underwater environment are still the open issues.

Motivated by the above considerations, in this work, we investigate the UAV-assisted multi-access computation offloading via hybrid NOMA and FDMA in marine networks. The main contributions of this work can be summarized as follows.

- We propose an energy efficient computation offloading scheme in marine networks, which consists of two transmission segments, i.e., the underwater acoustic transmission and the radio frequency transmission. In underwater acoustic transmission, multiple USNs collect sensing

data for data uploading via NOMA to improve the acoustic channel utilization. In radio frequency transmission, multiple UAVs hovering in the air provide aerial computing services as the aerial base stations, and USV offloads partial workloads to UAVs subject to the eavesdropping-attack from malicious node. We formulate a joint optimization problem of the USNs' uploading time, USV's computation offloading and offloading time, and the secrecy provisioning, with the objective of minimizing the total energy consumption of USNs and USV.

- Despite the non-convexity of the formulated joint optimization problem, we exploit a layered structure to decompose the problem. We divide the optimization problem into a top-problem for solving the USNs' uploading time and a sub-problem for optimizing the USV's computation offloading and offloading time, and the secrecy provisioning. Regarding the sub-problem, given the USNs' uploading time, USV's offloading time and the secrecy provisioning, we first solve the optimal computation offloading for USV. Then, we propose a two-dimensional line-searching algorithm to find the optimal USV's offloading time and the secrecy provisioning. Regarding the top-problem, we adopt the line-searching method to find the optimal USNs' uploading time.
- We finally provide simulations to evaluate the performance of the proposed algorithms. Numerical results demonstrate that the proposed algorithms can obtain the minimum energy consumption, and verify the effectiveness and efficiency of the proposed algorithms in comparison with the globally optimal solution by LINGO.

The remainder of this paper is organized as follows. Section II provides the literature review. Section III presents the system model and problem formulation. The transformation of the joint optimization problem is presented in Section IV. Section V proposes algorithms for solving the joint optimization problem. Numerical results are shown in Section VI. Section VII closes this paper with conclusion.

## II. LITERATURE REVIEW

This section reviews the related studies including (i) the joint computation offloading and resource allocation, and (ii) the energy efficient computation offloading in maritime networks.

Marine edge computing has been considered as a promising paradigm to improve the computation offloading efficiency in terms of energy consumption and computation latency in marine networks. There have been many studies exploiting the resource allocation for computation offloading in marine environment. In [22], Liu et al. investigated a two-layer UAVs maritime communication architecture to minimize the communication and computation latency for maritime UAV swarm mobile edge computing network. In [23], Abdel-Basset et al. proposed an energy-aware model based on the marine predators algorithm to implement the task scheduling in fog computing, with the objective of improving the quality of service (QoS) required by users. In [24], Su et al. exploited edge-based maritime networks services, in which the edge computation resource allocation is

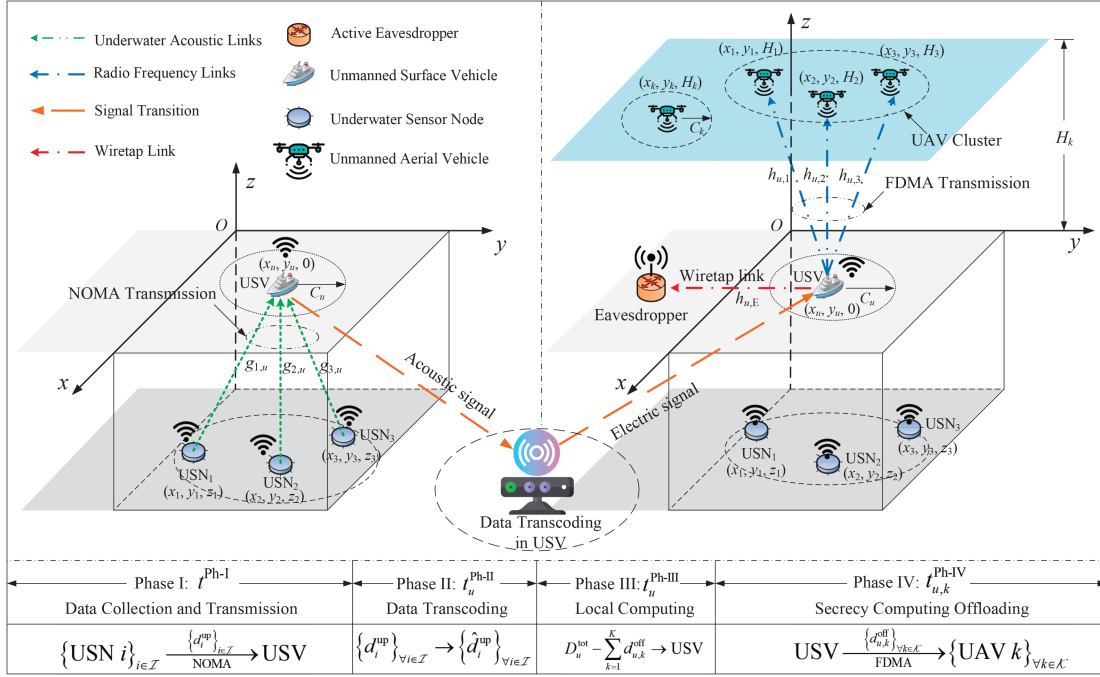


Fig. 1. UAV-assisted multi-access computation offloading via hybrid NOMA and FDMA in marine networks.

thoroughly analyzed. In [25], Liu et al. studied a spatio-temporal multi-graph convolutional network-based trajectory prediction scheme to guarantee traffic efficiency based on marine edge computing paradigm. On the other hand, fog computing has been adopted in marine communication networks, Yang et al. proposed the fog computing to Ocean of Things to reduce the amount of ocean data transmission and data processing delay [26]. In [27], a novel transmission scheduling scheme based on the enhanced deep Q-learning algorithm has been proposed in marine networks. In [28], Han et al. proposed a multi-AUV collaborative data collection algorithm based on Q-learning method to reduce the load of data collection task.

There have been many studies investigating the energy efficient computation offloading to prolong the devices' lifetime. In [29], a novel stochastic model has been proposed for opportunistic routing in multi-modal USNs, aiming to reduce the energy consumption and improve the network data delivery ratio. In [30], Timmins et al. studied the effect of sea surface shadowing conditions via the finite-difference time domain method on the marine communications channel. In [31], Li et al. investigated the maritime coverage enhancement using UAVs coordinated with hybrid satellite-terrestrial networks. Zeng et al. [32] established the systematic USV kinetics and information transmission models, jointly optimizing the trajectory and communication resource allocation to maximize the minimum expected throughput overall ships. In [33], a cooperative multicast communication scheme for maritime users relying on joint beamforming optimization and relay design has been proposed to increase the system throughput. In [34], an overview of a space-based network for the maritime surveillance information sharing has been provided to evaluate the coverage, throughput and real-time reception capacity of the presented systems. In [35], Yang et al. studied

the green energy and content-aware data transmissions in maritime networks. Different from existing works, in this work, we exploit the energy efficient computation offloading via hybrid NOMA and FDMA in marine networks. Specifically, we consider two transmission modes, i.e., in underwater segment, these USNs upload their sensing data to USV via NOMA; in radio frequency segment, USV aggregates the collected sensing data and then partially offloads it to UAVs via FDMA. We formulate a joint optimization problem to optimize the transmission time, the offloading workloads and the secrecy provisioning, with the objective of minimizing the total energy consumption of USNs and USV.

### III. SYSTEM MODEL AND PROBLEM FORMULATION

This section first introduces the system model for UAV-assisted multi-access computation offloading in marine networks, and the problem formulation is then proposed.

#### A. System Model

Fig. 1 illustrates the scenario of UAV-assisted multi-access computation offloading via hybrid NOMA and FDMA in marine networks, which consists of multiple USNs, one USV, multi-UAVs and an active eavesdropper. Specifically, in Phase I, the collected data from multi-USNs are uploaded to USV via NOMA. USV is responsible for collecting data from USNs through acoustic signal in underwater transmission and then transcodes data for radio frequency offloading to UAVs. In Phase II, USV receives the data from USNs through acoustic transmission and then transcodes the data for radio frequency transmission. In Phase III and Phase IV, USV processes partial workloads locally and then offloads the remained workloads to UAVs via FDMA while taking into

consideration that an eavesdropper overhears the USV's off-loading transmission.

*1) Phase I for Underwater Acoustic Transmission Between USNs and USV:* A group of USNs are deployed in the seabed to monitor marine environment. The set of USNs is denoted by  $\mathcal{I} = \{1, \dots, i, \dots, I\}$ . The collected data volume of USN  $i$  for uploading is  $\{d_i^{\text{up}}\}_{\forall i \in \mathcal{I}}$ , which requires transmitting data to USV. We use the subscript "u" to denote the USV, and the USV acts as the relay node to relay data between USNs and UAVs. In cartesian coordinate system, the position of USN  $i$  is expressed by  $(x_i, y_i, z_i)$ . The position of USV is denoted as  $(x_u, y_u, 0)$ , due to the fluidity of seawater, the USV will shift in a region of radius  $\mathcal{C}_u$ . The displacement variables of USV are denoted by  $x'_u$  and  $y'_u$ , which satisfy

$$\sqrt{(x'_u - x_u)^2 + (y'_u - y_u)^2} \leq \mathcal{C}_u. \quad (1)$$

The data transmission between USNs and USV is based on the underwater acoustic communication channels, according to [36], we can formulate the underwater acoustic communication model as

$$\mathcal{F}(\delta_{i,u}, \theta) = \delta_{i,u}^{\kappa} \mathcal{H}(\theta)^{\frac{\delta_{i,u}}{1000}}, \forall i \in \mathcal{I}, \quad (2)$$

where  $\mathcal{F}(\delta_{i,u}, \theta)$  denotes the attenuation of underwater acoustic signal. Parameter  $\theta$  is the central frequency of the acoustic signal.  $\delta_{i,u}$  represents the distance between USN  $i$  and USV. Parameter  $\kappa$  is a spreading factor.  $\mathcal{H}(\theta)$  denotes the absorption coefficient.

Based on the positions of USNs and USV, we can compute the distance between USN  $i$  and USV  $u$  as

$$\delta_{i,u} = \sqrt{(x_u - x_i)^2 + (y_u - y_i)^2 + z_i^2}, \forall i \in \mathcal{I}. \quad (3)$$

The absorption coefficient is obtained by the Thorp's empirical formula [36]

$$\mathcal{H}(\theta) = 0.11 \frac{\theta^2}{1 + \theta^2} + 44 \frac{\theta^2}{4100 + \theta^2} + 2.75e^{-4\theta^2} + 0.003. \quad (4)$$

Different from the traditional white noise, the underwater acoustic signals are generally affected by the oceanic noise during transmissions, which are usually divided into four major components: the turbulence  $\mathcal{R}_1(\theta)$ , the shipping  $\mathcal{R}_2(\theta)$ , the wind-driven waves  $\mathcal{R}_3(\theta)$  and the thermal noise  $\mathcal{R}_4(\theta)$  as follows according to [37]

$$\begin{cases} 10\log_{10} \mathcal{R}_1(\theta) = 17 - 30\log_{10} \theta, & (5a) \\ 10\log_{10} \mathcal{R}_2(\theta) = 40 + 20(a - 0.5) + 26\log_{10} \theta \\ \quad - 60\log_{10}(\theta + 0.03), & (5b) \\ 10\log_{10} \mathcal{R}_3(\theta) = 50 + 7.5b^{\frac{1}{2}} + 20\log_{10} \theta \\ \quad - 40\log_{10}(\theta + 0.4), & (5c) \\ 10\log_{10} \mathcal{R}_4(\theta) = -15 + 20\log_{10} \theta. & (5d) \end{cases}$$

Parameter  $a$  in (5b) denotes the medium shipping activity factor, and parameter  $b$  in (5c) represents the wind speed. The

total noise power density can be calculated by  $\mathcal{R}(\theta) = \mathcal{R}_1(\theta) + \mathcal{R}_2(\theta) + \mathcal{R}_3(\theta) + \mathcal{R}_4(\theta)$ .

According to [37], the underwater acoustic channel gain between USN  $i$  and USV  $u$  can be computed as

$$g_{i,u} = \frac{1}{\mathcal{F}(\delta_{i,u}, \theta) \mathcal{R}(\theta) W_i}, \forall i \in \mathcal{I}, \quad (6)$$

where parameter  $W_i$  denotes the channel bandwidth of USN  $i$  for underwater acoustic transmission.  $\mathcal{R}(\theta)$  is the ocean noise power such as the turbulence, the shipping, the wind-driven waves and the thermal noise.

In a NOMA cluster of USNs, we assume that the channel gains of USNs are ordered by

$$g_{1,u} > g_{2,u} > \dots > g_{i,u} > g_{j,u} > \dots > g_{I,u}. \quad (7)$$

Thanks to the successive interference cancellation (SIC) technique, SIC enables a group of USNs to simultaneously upload their data to USV. In underwater acoustic link, the received signal to noise plus interference (SINR) of USN  $i$ ,  $\forall i \in \mathcal{I}$  can be computed by

$$\psi_{i,u}^{\text{Ph-I}} = \frac{p_{i,u} g_{i,u}}{\sum_{j \in \mathcal{I}, j < i} p_{j,u} g_{j,u} + 1}, \forall i \in \mathcal{I}, \quad (8)$$

where parameter  $p_{i,u}$  denotes the transmission power of USN  $i$  to USV  $u$ , parameter  $g_{i,u}$  is the channel gain from USN  $i$  to USV  $u$ . According to the Shannon formula, the achievable transmission rate of USN  $i$ ,  $\forall i \in \mathcal{I}$  can be computed by

$$r_{i,u}^{\text{Ph-I}} = W_i \log_2 (1 + \psi_{i,u}^{\text{Ph-I}}), \forall i \in \mathcal{I}. \quad (9)$$

In a NOMA cluster, the required power of USN  $i$  for data transmission can be calculated by

$$p_{i,u}^{\text{Ph-I}} \left( \left\{ \psi_{j,u}^{\text{Ph-I}} \right\}_{j \in \mathcal{I}, j < i} \right) = \frac{1}{g_{i,u}} \psi_{i,u}^{\text{Ph-I}} \prod_{j \in \mathcal{I}, j < i} (1 + \psi_{j,u}^{\text{Ph-I}}), \quad \forall i \in \mathcal{I}. \quad (10)$$

Let  $t^{\text{Ph-I}}$  denote the uploading time of USNs transmitting data to USV  $u$ , it can be computed by

$$\frac{d_i^{\text{up}}}{t^{\text{Ph-I}}} = W_i \log_2 (1 + \psi_{i,u}^{\text{Ph-I}}), \forall i \in \mathcal{I}. \quad (11)$$

Based on (11), we can rewrite the received SINR of USN  $i$  as follows

$$\psi_{i,u}^{\text{Ph-I}} = 2^{\frac{d_i^{\text{up}}}{t^{\text{Ph-I}} W_i}} - 1, \forall i \in \mathcal{I}. \quad (12)$$

Substituting (12) into (10), the required power of USN  $i$  for data transmission is [38]

$$p_{i,u}^{\text{Ph-I}} \left( \left\{ \psi_{j,u}^{\text{Ph-I}} \right\}_{j \in \mathcal{I}, j < i} \right) = \frac{1}{g_{i,u}} \left( 2^{\frac{d_i^{\text{up}}}{t^{\text{Ph-I}} W_i}} - 1 \right) 2^{\frac{1}{t^{\text{Ph-I}} W_i} \sum_{j \in \mathcal{I}, j < i} d_j^{\text{up}}}, \forall i \in \mathcal{I}. \quad (13)$$

The energy consumption for uploading data  $d_i^{\text{up}}$  to USV can be expressed as

$$E_i^{\text{Ph-I}} = p_{i,u}^{\text{Ph-I}} t^{\text{Ph-I}}, \forall i \in \mathcal{I}. \quad (14)$$

2) *Phase II for Data Transcoding of USV*: USV first receives the data  $d_i^{\text{up}}$  from USNs through acoustic transmission and then transcodes the data for radio frequency transmission. The total amount of transcoded data by USV can be expressed as

$$D_u^{\text{tot}} = \sum_{i=1}^I d_i^{\text{up}}. \quad (15)$$

The transcoding time can be expressed as

$$t_u^{\text{Ph-II}} = \frac{v_u D_u^{\text{tot}}}{\rho_u}, \quad (16)$$

where parameter  $v_u$  indicates the number of CPU cycles for processing the conversion of one bit of acoustic signal. Parameter  $\rho_u$  denotes the processing capability of USV in CPU cycles per second. The energy consumption for transcoding data  $D_u^{\text{tot}}$  can be expressed as [37]

$$E_u^{\text{Ph-II}} = \epsilon_u \rho_u^3 t_u^{\text{Ph-II}} = \epsilon_u \rho_u^2 v_u D_u^{\text{tot}}, \quad (17)$$

where parameter  $\epsilon_u$  is the power consumption coefficient of USV  $u$ .

3) *Phase III for Local Computing of USV*: Let  $d_{u,k}^{\text{off}}$  denote the required offloading data of USV  $u$  to UAV  $k$ , the local computing time for USV  $u$  can be expressed as

$$t_u^{\text{Ph-III}} = \frac{v_u \left( D_u^{\text{tot}} - \sum_{k=1}^K d_{u,k}^{\text{off}} \right)}{\rho_u}. \quad (18)$$

Therefore, the energy consumption of USV  $u$  for local computing can be expressed as

$$E_u^{\text{Ph-III}} = \epsilon_u \rho_u^3 t_u^{\text{Ph-III}} = \epsilon_u \rho_u^2 v_u \left( D_u^{\text{tot}} - \sum_{k=1}^K d_{u,k}^{\text{off}} \right). \quad (19)$$

4) *Phase IV for Secrecy Computation Offloading Between USV and UAVs*: We consider a multi-access computation offloading that USV offloads data to a cluster of UAVs. The set of UAVs flying in the air is denoted by  $\mathcal{K} = \{1, \dots, k, \dots, K\}$ . UAVs hovering at a fixed height act as the aerial servers to perform data computing. The position of UAV  $k$  is expressed as  $(x_k, y_k, H_k)$ , where  $H_k$  is the fixed height of UAV  $k$ . Considering the flexibility and the fluidity of wind, UAV  $k$  shifts in a

region of radius  $C_k$ . The displacement variables of UAV  $k$  are denoted by  $x'_k$  and  $y'_k$ , which satisfy

$$\sqrt{(x'_k - x_k)^2 + (y'_k - y_k)^2} \leq C_k, \forall k \in \mathcal{K}. \quad (20)$$

We consider that USV adopts the FDMA mode to offload data to UAVs. The data transmission between USV and UAV follows the line-of-sight (LoS) propagation, and the channel power gain between USV  $u$  and UAV  $k$  can be expressed by

$$h_{u,k} = \frac{\chi}{\left[ (x_u - x_k)^2 + (y_u - y_k)^2 + H_k^2 \right]^\eta}, \forall k \in \mathcal{K}, \quad (21)$$

where parameter  $\chi$  is the channel power gain associated with a reference distance such as 1 m. Parameter  $\eta$  is the path loss exponent which is higher than 1. The data rate of the uplink transmission between USV  $v$  and UAV  $k$  can be expressed as

$$r_{u,k}^{\text{Ph-IV}} = W_{u,k} \log_2 \left( 1 + \frac{q_{u,k} h_{u,k}}{N_u} \right), \forall k \in \mathcal{K}, \quad (22)$$

where parameter  $W_{u,k}$  is the channel bandwidth allocated to UAV  $k$ . Parameter  $q_{u,k}$  denotes the transmission power between USV  $u$  and UAV  $k$ . Parameter  $N_u$  represents the powers of background noise at USV.

As shown in Fig. 1, the active eavesdropper can overhear the offloaded data of USV. Let  $h_{u,E}$  denote the channel power gain between USV  $u$  and the eavesdropper, the data rate at the eavesdropper for eavesdropping can be expressed as

$$r_{u,E}^{\text{Ph-IV}} = W_{u,k} \log_2 \left( 1 + \frac{q_{u,k} h_{u,E}}{N_E} \right), \quad (23)$$

where parameter  $N_E$  represents the power of background noise at the eavesdropper. The secrecy capacity of USV  $u$  associated with UAV  $k$  can be expressed as the difference between the channel capacity of the uplink transmission and that of the wiretap channel, i.e.,

$$C_{u,k} = \left[ r_{u,k}^{\text{Ph-IV}} - r_{u,E}^{\text{Ph-IV}} \right]^+, \forall k \in \mathcal{K}. \quad (24)$$

where  $[\times]^+ = \max\{\times, 0\}$ . Since the eavesdropper may intentionally hide its information (e.g., location, channel gain), the value of  $h_{u,E}$  cannot be obtained. Similar to [39], we consider the secrecy-outage probability (SOP) to estimate the performance of eavesdropping when offloading data to UAV. Here, the SOP is defined as the probability that the secrecy capacity  $C_{u,k}$  is smaller than an allocated threshold secrecy capacity  $r_{u,k}$ . Therefore, we can express the SOP as a function of the transmission power  $q_{u,k}$  and the allocated data rate  $r_{u,k}$  as follows

$$\begin{aligned} P_{u,k}(q_{u,k}, r_{u,k}) &= 1 - \Pr \left\{ C_{u,k} \geq r_{u,k} \mid W_{u,k} \log_2 \left( 1 + \frac{q_{u,k} h_{u,k}}{N_u} \right) \right. \\ &\quad \left. \geq W_{u,k} \log_2 \left( 1 + \frac{q_{u,k} h_{u,E}}{N_E} \right) \right\}, \forall k \in \mathcal{K}, \end{aligned} \quad (25)$$

where the conditional probability means that USV will upload its data to UAV only if it can achieve a nonnegative secrecy-rate.

Assume that the value of  $h_{u,E}$  follows the exponential distribution with the mean of  $\mu_{u,E}$ . Here, the value of  $\mu_{u,E}$  indicates the average eavesdropping-path strength. Therefore, the SOP can be rewritten as

$$P_{u,k}(q_{u,k}, r_{u,k}) = 1 - \Pr\left\{h_{u,E} \leq \frac{N_E}{q_{u,k}} \left(2^{-\frac{r_{u,k}}{W_{u,k}}} - 1\right) + \hat{h}_{u,k} 2^{-\frac{r_{u,k}}{W_{u,k}}} | h_{u,E} \leq \hat{h}_{u,k}\right\}, \forall k \in \mathcal{K}. \quad (26)$$

We introduce an auxiliary parameter  $\hat{h}_{u,k} = \frac{N_E}{N_u} h_{u,k}$ , which can be explained as the effective channel gain from USV  $u$  to UAV  $k$ . According to the conditional probability formula and manipulations, we can obtain the SOP when USV transmits data to UAV as follows:

$$P_{u,k}(q_{u,k}, r_{u,k}) = \frac{1}{1 - e^{-\frac{\hat{h}_{u,k}}{\mu_{u,E}}}} \times \left\{ e^{-\frac{\frac{N_E}{q_{u,k}} \left(2^{-\frac{r_{u,k}}{W_{u,k}}} - 1\right) + \frac{N_E}{N_u} \hat{h}_{u,k} 2^{-\frac{r_{u,k}}{W_{u,k}}}}{\mu_{u,E}}} - e^{-\frac{\hat{h}_{u,k}}{\mu_{u,E}}} \right\}, \forall k \in \mathcal{K}. \quad (27)$$

The SOP provides an approach for measuring the secrecy-level when USV uploads data to UAV. In specific, at the given values of transmission power  $q_{u,k}$  and data rate  $r_{u,k}$ , the SOP means the probability of USV failing to protect the data from being overheard by the eavesdropper. In this sense, the low value of SOP can make a high secrecy-level for data transmission. In particular, when there is no data transmission between USV and UAV (i.e.,  $r_{u,k} = 0$ ), we have  $P_{u,k}(q_{u,k}, r_{u,k}) = 0$ . The result also validates the feasibility of SOP for measuring the secrecy-level.

We introduce a variable  $\varsigma_k \in [0, 1]$  to satisfy the requirement of secrecy-outage when transmitting data, which means that the SOP should be no larger than a given value of  $\varsigma_k$ . This leads to the following constraint

$$P_{u,k}(q_{u,k}, r_{u,k}) \leq \varsigma_k, \forall k \in \mathcal{K}. \quad (28)$$

The uploading time of USV  $u$  offloading data to UAV  $k$  can be expressed as

$$t_{u,k}^{\text{Ph-IV}} = \frac{d_{u,k}^{\text{off}}}{r_{u,k}(1 - \varsigma_k)}, \forall k \in \mathcal{K}. \quad (29)$$

Therefore, the energy consumption of USV  $u$  for offloading data can be expressed as

$$E_u^{\text{Ph-IV}} = \sum_{k=1}^K t_{u,k}^{\text{Ph-IV}} q_{u,k}. \quad (30)$$

The overall latency to complete USNs' computation workloads can be expressed as

$$t^{\text{ove}} = t^{\text{Ph-I}} + t_u^{\text{Ph-II}} + \max\left\{t_u^{\text{Ph-III}}, t_{u,k}^{\text{Ph-IV}} + \max_{k \in \mathcal{K}} \left\{ \frac{v_k d_{u,k}^{\text{off}}}{\rho_k} \right\} \right\}, \quad (31)$$

where parameter  $v_k$  indicates the number of CPU cycles of UAV  $k$  for processing one bit data. Parameter  $\rho_k$  denotes the processing capability of UAV  $k$  in CPU cycles per second. Specifically, the overall latency  $t^{\text{ove}}$  can be explained as follows: the NOMA uploading time  $t^{\text{Ph-I}}$  in Phase I, the transcoding time  $t_u^{\text{Ph-II}}$  in Phase II, the local computing time  $t_u^{\text{Ph-III}}$  of USV  $u$  in Phase III, the FDMA uploading time  $t_{u,k}^{\text{Ph-IV}}$  and the UAV's computing latency  $\frac{v_k d_{u,k}^{\text{off}}}{\rho_k}$  in Phase IV.

### B. Problem Formulation

Based on the above modelings, the total energy consumption of USNs and USV can be expressed as

$$E^{\text{tot}} = \underbrace{\sum_{i=1}^I E_i^{\text{Ph-I}}}_{\text{USNs}} + \underbrace{E_u^{\text{Ph-II}} + E_u^{\text{Ph-III}} + E_u^{\text{Ph-IV}}}_{\text{USV}}. \quad (32)$$

We aim to minimize the total energy consumption for USV and USNs, in which the energy consumption of USNs consists of the uploading energy of the total amount of data in Phase I, the energy consumption of USV is comprised of the data transcoding energy in Phase II, the local computing energy in Phase III and the computation offloading energy in Phase IV. We formulate a joint optimization for the USN's uploading time  $t^{\text{Ph-I}}$ , the USV's computation offloading  $d_{u,k}^{\text{off}}$ , the offloading time  $t_{u,k}^{\text{Ph-IV}}$  and the secrecy provisioning  $\varsigma_k$ , with the objective of minimizing the total energy consumption of USNs and USV. The formulated Energy Consumption Minimization (ECM) is as follows

$$(\text{ECM}) : E_{\text{tot}}^{\min} = \min E^{\text{tot}}$$

$$\text{subject to : } 0 \leq d_i^{\text{up}} \leq D_i^{\max}, \forall i \in \mathcal{I}, \quad (33)$$

$$0 \leq p_{i,u}^{\text{Ph-I}} \leq P_{i,u}^{\max}, \forall i \in \mathcal{I}, \quad (34)$$

$$0 \leq q_{u,k} \leq Q_{u,k}^{\max}, \forall k \in \mathcal{K}, \quad (35)$$

$$0 \leq d_{u,k}^{\text{off}} \leq D_{u,k}^{\text{tot}}, \forall k \in \mathcal{K}, \quad (36)$$

$$0 \leq \sum_{k=1}^K d_{u,k}^{\text{off}} \leq D_u^{\text{tot}}, \quad (37)$$

$$0 \leq t^{\text{Ph-I}} \leq t^{\max}, \quad (38)$$

$$0 \leq t^{\text{ove}} \leq T^{\max}, \quad (39)$$

constraints (1), (20) and (28)

variables :  $t^{\text{Ph-I}}, \{d_{u,k}^{\text{off}}\}_{\forall k \in \mathcal{K}}, \{t_{u,k}^{\text{Ph-IV}}\}_{\forall k \in \mathcal{K}}$ , and  $\{\varsigma_k\}_{\forall k \in \mathcal{K}}$ .

Constraint (33) ensures that the uploading data of USN  $i$  cannot exceed its maximum data volume  $D_i^{\max}$  due to the limited storage capacity. Considering the limited power capacity of USN and USV, constraint (34) and constraint (35) mean that USN  $i$ 's transmission power and USV  $u$ 's transmission

power cannot exceed the maximum  $P_{i,u}^{\max}$  and  $Q_{u,k}^{\max}$ , respectively. Constraint (36) guarantees that the offloaded data to each UAV cannot exceed the total amount of data  $D_u^{\text{tot}}$ . Constraint (37) ensures that the total amount of offloaded data to multi-UAVs cannot exceed  $D_u^{\text{tot}}$ . Constraint (38) means that the NOMA transmission latency in Phase I cannot exceed the delay-limitation  $t^{\max}$ . Constraint (39) means that the overall delay for completing USNs' workloads cannot exceed the delay-limitation  $T^{\max}$ .

#### IV. TRANSFORMATION OF THE JOINT OPTIMIZATION PROBLEM

It can be identified that the objective function of Problem (ECM) is strictly nonconvex, and it is computational challenging to solve Problem (ECM) [40]. In this section, we first find an equivalent layered form of Problem (ECM) and propose a layered approach to solve it.

##### A. Equivalent Form of Problem (ECM)

Based on (27) and (28), we obtain

$$\frac{N_E}{q_{u,k}} \left( 2^{-\frac{r_{u,k}}{W_{u,k}}} - 1 \right) + \frac{N_E}{N_u} h_{u,k} 2^{-\frac{r_{u,k}}{W_{u,k}}} \geq -\mu_{u,E} \ln \left( e^{-\frac{\hat{h}_{u,k}}{\mu_{u,E}}} (1 - \varsigma_k) + \varsigma_k \right). \quad (40)$$

We introduce an auxiliary parameter  $v_{u,k}$ , which is given by

$$v_{u,k} = -\mu_{u,E} \ln \left( e^{-\frac{\hat{h}_{u,k}}{\mu_{u,E}}} (1 - \varsigma_k) + \varsigma_k \right), \forall k \in \mathcal{K}. \quad (41)$$

Notice that parameter  $v_{u,k}$  is affected by the effective channel gain  $\hat{h}_{u,k}$ , the average eavesdropping-path strength  $\mu_{u,E}$  and the limitation of secrecy-outage  $\varsigma_k$ , which quantifies the influence of security requirement.

Based on (40) and (41), the allocated data rate  $r_{u,k}$  has the following constraint, which is essentially equivalent to (28):

$$r_{u,k} \leq W_{u,k} \log_2 \left( \frac{q_{u,k} \hat{h}_{u,k} + N_E}{q_{u,k} v_{u,k} + N_E} \right), \forall k \in \mathcal{K}. \quad (42)$$

A keen observation on (42) is that  $\hat{h}_{u,k} > v_{u,k}$  to guarantee a positive offloading rate. In addition, the right hand of (42) is increasing with the transmission power  $q_{u,k}$ . Therefore, at the optimal offloading rate of Problem (ECM), constraint (42) is strictly binding. We obtain

$$r_{u,k} = W_{u,k} \log_2 \left( \frac{q_{u,k} \hat{h}_{u,k} + N_E}{q_{u,k} v_{u,k} + N_E} \right), \forall k \in \mathcal{K}. \quad (43)$$

Through a simple transformation, the transmission power  $q_{u,k}$  of USV  $u$  for offloading can be expressed as

$$q_{u,k} = \frac{N_E \left( 2^{\frac{r_{u,k}}{W_{u,k}}} - 1 \right)}{\hat{h}_{u,k} - v_{u,k} 2^{\frac{r_{u,k}}{W_{u,k}}}}, \forall k \in \mathcal{K}. \quad (44)$$

Based on (29), the allocated data rate  $r_{u,k}$  can be expressed as

$$r_{u,k} = \frac{d_{u,k}^{\text{off}}}{t_{u,k}^{\text{Ph-IV}} (1 - \varsigma_k)}, \forall k \in \mathcal{K}. \quad (45)$$

Therefore, the transmission power  $q_{u,k}$  of USV  $u$  can be rewritten as

$$q_{u,k} = \frac{N_E \left( 2^{\frac{d_{u,k}^{\text{off}}}{W_{u,k} t_{u,k}^{\text{Ph-IV}} (1 - \varsigma_k)}} - 1 \right)}{\hat{h}_{u,k} - v_{u,k} 2^{\frac{d_{u,k}^{\text{off}}}{W_{u,k} t_{u,k}^{\text{Ph-IV}} (1 - \varsigma_k)}}}, \forall k \in \mathcal{K}. \quad (46)$$

Based on the above operations, we obtain the following equivalent form of Problem (ECM):

$$\begin{aligned} \text{(ECM-E)} : E_{\text{tot}}^{\min} = \min & \sum_{i=1}^I t^{\text{Ph-I}} p_{i,u}^{\text{Ph-I}} + \epsilon_u v_u \rho_u^2 \sum_{i=1}^I d_i^{\text{up}} \\ & + \epsilon_u v_u \rho_u^2 \left( D_u^{\text{tot}} - \sum_{k=1}^K d_{u,k}^{\text{off}} \right) + \sum_{k=1}^K t_{u,k}^{\text{Ph-IV}} q_{u,k} \end{aligned}$$

subject to :constraints (31), (32), (33), (34), (35), (36), and (37)

variables :  $t^{\text{Ph-I}}$ ,  $\{d_{u,k}^{\text{off}}\}_{\forall k \in \mathcal{K}}$ ,  $\{t_{u,k}^{\text{Ph-IV}}\}_{\forall k \in \mathcal{K}}$ , and  $\{\varsigma_k\}_{\forall k \in \mathcal{K}}$ .

##### B. Decomposition of Problem (ECM)

Problem (ECM-E) is still a strictly nonconvex optimization problem. To solve it efficiently, we propose the following decomposition approach.

1) Sub-problem under given  $t^{\text{Ph-I}}$ : Suppose that  $t^{\text{Ph-I}}$  is given in advance. We first solve the following sub-problem:

$$\begin{aligned} \text{(ECM-E-Sub)} : E_{\text{tot-sub}}^{\min} = \min & \sum_{i=1}^I t^{\text{Ph-I}} p_{i,u}^{\text{Ph-I}} + \epsilon_u v_u \rho_u^2 \sum_{i=1}^I d_i^{\text{up}} \\ & + \epsilon_u v_u \rho_u^2 \left( D_u^{\text{tot}} - \sum_{k=1}^K d_{u,k}^{\text{off}} \right) + \sum_{k=1}^K t_{u,k}^{\text{Ph-IV}} q_{u,k} \end{aligned}$$

subject to :constraints (31), (32), (33), (34), (35), and (37),

variables :  $\{d_{u,k}^{\text{off}}\}_{\forall k \in \mathcal{K}}$ ,  $\{t_{u,k}^{\text{Ph-IV}}\}_{\forall k \in \mathcal{K}}$ , and  $\{\varsigma_k\}_{\forall k \in \mathcal{K}}$ .

2) Top-problem to optimize  $t^{\text{Ph-I}}$ : After solving Problem (ECM-E-Sub) and obtaining the value of  $E_{\text{tot-sub}}^{\min}$ , we then continue to optimize the value of  $t^{\text{Ph-I}}$ , which corresponds to the following optimization problem:

$$\begin{aligned}
(\text{ECM-E-Top}) : \quad & E_{\text{tot-top}}^{\min} = \min E_{\text{tot-top}}^{\min} \\
\text{subject to :} \quad & \text{constraint (36),} \\
\text{variables :} \quad & t^{\text{Ph-I}}.
\end{aligned}$$

### C. Layer Structure of Problem (ECM-E-Sub)

A key observation on Problem (ECM-E-Sub) is that  $q_{u,k}$  in (46) is strictly convex with respect to  $d_{u,k}^{\text{off}}$  when  $t_{u,k}^{\text{Ph-IV}}$  and  $\varsigma_k$  are given. Exploiting this important feature, we propose the following decomposition of Problem (ECM-E).

Based on (31), we obtain the constraint for the required off-loading data

$$D_u^{\text{tot}} - \frac{\rho_u(T^{\text{max}} - t^{\text{Ph-I}} - t_u^{\text{Ph-II}})}{v_u} \leq \sum_{k=1}^K d_{u,k}^{\text{off}} \leq D_u^{\text{tot}}. \quad (47)$$

1) Sub-problem to optimize  $d_{u,k}^{\text{off}}$  under given  $t_{u,k}^{\text{Ph-IV}}$  and  $\varsigma_k$ : Given  $t_{u,k}^{\text{Ph-IV}}$  and  $\varsigma_k$ , Problem (ECM-E-Sub) turns into a sub-problem for optimizing  $d_{u,k}^{\text{off}}$  as follows

$$\begin{aligned}
(\text{ECM-}\hat{\text{E}}\text{-Sub}) : \quad & \hat{E}_{\text{tot-sub}}^{\min} = \min \sum_{i=1}^I t^{\text{Ph-I}} p_{i,u}^{\text{Ph-I}} + \epsilon_u v_u \rho_u^2 \sum_{i=1}^I d_i^{\text{up}} \\
& + \epsilon_u v_u \rho_u^2 \left( D_u^{\text{tot}} - \sum_{k=1}^K d_{u,k}^{\text{off}} \right) + \sum_{k=1}^K t_{u,k}^{\text{Ph-IV}} q_{u,k} \\
\text{subject to :} \quad & \text{constraint (47),} \\
\text{variables :} \quad & \{d_{u,k}^{\text{off}}\}_{\forall k \in \mathcal{K}}.
\end{aligned}$$

We use  $\hat{E}_{\text{tot-sub}}^{\min}$  to denote the optimal value, i.e., the minimum energy consumption of USNs and USV under the given tuple of  $t_{u,k}^{\text{Ph-IV}}$  and  $\varsigma_k$ .

2) Top-problem to optimize  $t_{u,k}^{\text{Ph-IV}}$  and  $\varsigma_k$ : With the optimal value of  $\hat{E}_{\text{tot-sub}}^{\min}$  from solving Problem (ECM- $\hat{\text{E}}$ -Sub), we then solve the following top-problem that jointly optimizes  $t_{u,k}^{\text{Ph-IV}}$  and  $\varsigma_k$ :

$$\begin{aligned}
(\text{ECM-}\hat{\text{E}}\text{-Top}) : \quad & \hat{E}_{\text{tot-top}}^{\min} = \min \hat{E}_{\text{tot-sub}}^{\min} \\
\text{subject to :} \quad & 0 \leq t_{u,k}^{\text{Ph-IV}} \leq T^{\text{max}} - t^{\text{Ph-I}} - t_u^{\text{Ph-II}}, \quad (48) \\
& \text{constraint (26),} \\
\text{variables :} \quad & \{t_{u,k}^{\text{Ph-IV}}\}_{\forall k \in \mathcal{K}} \text{ and } \{\varsigma_k\}_{\forall k \in \mathcal{K}}.
\end{aligned}$$

Constraint (48) comes from the constraint (39), which is obtained by relaxing the UAV's computing delay.

With the above decomposition, we exploit efficient algorithm to solve Problem (ECM). The details are shown in the next section.

## V. PROPOSED ALGORITHM FOR SOLVING THE JOINT OPTIMIZATION PROBLEM

### A. Analysis of Problem (ECM- $\hat{\text{E}}$ -Sub)

*Proposition 1:* Given  $t_{u,k}^{\text{Ph-IV}}$  and  $\varsigma_k$ , Problem (ECM- $\hat{\text{E}}$ -Sub) is a strictly convex optimization problem.

*Proof:* In Problem (ECM- $\hat{\text{E}}$ -Sub), (i) it can be identified that the third item, i.e.,  $\epsilon_u v_u \rho_u^2 (D_u^{\text{tot}} - \sum_{k=1}^K d_{u,k}^{\text{off}})$ , is strictly convex with respect to  $\{d_{u,k}^{\text{off}}\}_{\forall k \in \mathcal{K}}$ . When the value of  $t_{u,k}^{\text{Ph-IV}}$  is given in advance, an important feature is that the fourth item, i.e.,  $\sum_{k=1}^K t_{u,k}^{\text{Ph-IV}} q_{u,k}$ , is strictly convex with respect to  $\{d_{u,k}^{\text{off}}\}_{\forall k \in \mathcal{K}}$ .

(ii) Based on (46), the second derivative of  $q_{u,k}$  with respect to  $\{d_{u,k}^{\text{off}}\}_{\forall k \in \mathcal{K}}$  can be obtained by

$$\frac{\partial^2 q_{u,k}}{\partial d_{u,k}^{\text{off}^2}} = \frac{N_E (\hat{h}_{u,k} - v_{u,k}) 2^{\frac{d_{u,k}^{\text{off}}}{W_{u,k} t_{u,k}^{\text{Ph-IV}} (1-\varsigma_k)}} (\ln 2)^2 \mathcal{M}}{\left( W_{u,k} t_{u,k}^{\text{Ph-IV}} (1-\varsigma_k) \right)^2 \mathcal{N}}, \quad \forall k \in \mathcal{K}, \quad (49)$$

$$\begin{cases} \mathcal{M} = \hat{h}_{u,k} + v_{u,k} 2^{\frac{d_{u,k}^{\text{off}}}{W_{u,k} t_{u,k}^{\text{Ph-IV}} (1-\varsigma_k)}}, \\ \mathcal{N} = \left( \hat{h}_{u,k} - v_{u,k} 2^{\frac{d_{u,k}^{\text{off}}}{W_{u,k} t_{u,k}^{\text{Ph-IV}} (1-\varsigma_k)}} \right)^3, \end{cases}$$

where  $\mathcal{M}$  and  $\mathcal{N}$  are the auxiliary variables, and the values of  $\mathcal{M}$  and  $\mathcal{N}$  are higher than zero. It can be identified that the value of second derivative of  $q_{u,k}$  with respect to  $\{d_{u,k}^{\text{off}}\}_{\forall k \in \mathcal{K}}$  is higher than zero. Therefore, we can identify that  $q_{u,k}$  is strictly convex with respect to  $\{d_{u,k}^{\text{off}}\}_{\forall k \in \mathcal{K}}$  when the values of  $t_{u,k}^{\text{Ph-IV}}$  and  $\varsigma_k$  are fixed in advance. As a result, it can be concluded that Problem (ECM- $\hat{\text{E}}$ -Sub) is a strictly convex optimization problem with respect to  $\{d_{u,k}^{\text{off}}\}_{\forall k \in \mathcal{K}}$ . ■

Proposition 1 enables us to solve Problem (ECM- $\hat{\text{E}}$ -Sub) using the Karush-Kuhn-Tucker (KKT) conditions. By introducing dual variables  $\alpha$  and  $\beta$ , we obtain the Lagrangian function  $\mathcal{L}(d_{u,k}^{\text{off}}, \alpha, \beta)$  as follows

$$\begin{aligned}
\mathcal{L}(d_{u,k}^{\text{off}}, \alpha, \beta) = & \sum_{i=1}^I t^{\text{Ph-I}} p_{i,u}^{\text{Ph-I}} + \epsilon_u v_u \rho_u^2 \sum_{i=1}^I d_i^{\text{up}} + \epsilon_u v_u \rho_u^2 \left( D_u^{\text{tot}} - \sum_{k=1}^K d_{u,k}^{\text{off}} \right) \\
& + \sum_{k=1}^K t_{u,k}^{\text{Ph-IV}} q_{u,k} + \alpha \left\{ D_u^{\text{tot}} - \frac{\rho_u(T^{\text{max}} - t^{\text{Ph-I}} - t_u^{\text{Ph-II}})}{v_u} \right. \\
& \left. - \sum_{k=1}^K d_{u,k}^{\text{off}} \right\} + \beta \left( \sum_{k=1}^K d_{u,k}^{\text{off}} - D_u^{\text{tot}} \right). \quad (50)
\end{aligned}$$

We obtain the first derivative of  $\mathcal{L}(d_{u,k}^{\text{off}}, \alpha, \beta)$  with respect to  $d_{u,k}^{\text{off}}$  as follows

$$\begin{aligned}
\frac{\partial \mathcal{L}(d_{u,k}^{\text{off}}, \alpha, \beta)}{\partial d_{u,k}^{\text{off}}} = & -\epsilon_u v_u \rho_u^2 - \alpha + \beta \\
& + \frac{N_E (\hat{h}_{u,k} - v_{u,k}) 2^{\frac{d_{u,k}^{\text{off}}}{W_{u,k} t_{u,k}^{\text{Ph-IV}} (1-\varsigma_k)}} \ln 2}{\left( W_{u,k} (1-\varsigma_k) \left( \hat{h}_{u,k} - v_{u,k} 2^{\frac{d_{u,k}^{\text{off}}}{W_{u,k} t_{u,k}^{\text{Ph-IV}} (1-\varsigma_k)}} \right) \right)^2}. \quad (51)
\end{aligned}$$

Let (51) equal to zero, we obtain the following equation:

$$\frac{N_E(\hat{h}_{u,k} - v_{u,k})f(d_{u,k}^{\text{off}}) \ln 2}{W_{u,k}(1 - \varsigma_k)(\hat{h}_{u,k} - v_{u,k}f(d_{u,k}^{\text{off}}))^2} = \mathcal{F}(\alpha, \beta). \quad (52)$$

$f(d_{u,k}^{\text{off}})$  and  $\mathcal{F}(\alpha, \beta)$  are the auxiliary variables, which are defined as

$$f(d_{u,k}^{\text{off}}) = 2 \frac{d_{u,k}^{\text{off}}}{W_{u,k} t_{u,k}^{\text{Ph-IV}} (1 - \varsigma_k)}. \quad (53)$$

$$\mathcal{F}(\alpha, \beta) = \epsilon_u v_u \rho_u^2 + \alpha - \beta. \quad (54)$$

To ensure  $\mathcal{F}(\alpha, \beta) > 0$ , we set the upper bound as  $\beta^{\text{ub}} = \epsilon_u v_u \rho_u^2$ , the lower bound as  $\alpha^{\text{lb}} = -\epsilon_u v_u \rho_u^2$ . Through a simple transformation, (52) can be rewritten as the following quadratic equation:

$$\mathcal{A}(\alpha, \beta) f^2(d_{u,k}^{\text{off}}) - \mathcal{B}(\alpha, \beta) f(d_{u,k}^{\text{off}}) + \mathcal{C}(\alpha, \beta) = 0, \quad (55)$$

where the auxiliary variables  $\mathcal{A}$ ,  $\mathcal{B}$  and  $\mathcal{C}$  are given as

$$\begin{cases} \mathcal{A}(\alpha, \beta) = W_{u,k}(1 - \varsigma_k) v_{u,k}^2 \mathcal{F}(\alpha, \beta), \\ \mathcal{B}(\alpha, \beta) = 2 v_{u,k} \hat{h}_{u,k} W_{u,k}(1 - \varsigma_k) \mathcal{F}(\alpha, \beta) \\ \quad + N_E(\hat{h}_{u,k} - v_{u,k}) \ln 2, \\ \mathcal{C}(\alpha, \beta) = W_{u,k}(1 - \varsigma_k) \hat{h}_{u,k}^2 \mathcal{F}(\alpha, \beta). \end{cases} \quad (56a) \quad (56b) \quad (56c)$$

We compute the discriminant of the quadratic equation as

$$\begin{aligned} \Delta &= \mathcal{B}^2(\alpha, \beta) - 4\mathcal{A}(\alpha, \beta)\mathcal{C}(\alpha, \beta) \\ &= 4 v_{u,k} \hat{h}_{u,k} \mathcal{F}(\alpha, \beta) W_{u,k}(1 - \varsigma_k) N_E(\hat{h}_{u,k} - v_{u,k}) \ln 2 \\ &\quad + N_E^2(\hat{h}_{u,k} - v_{u,k})^2 (\ln 2)^2. \end{aligned} \quad (57)$$

Due to  $0 < \varsigma_k < 1$  and  $\hat{h}_{u,k} > v_{u,k}$ , we obtain  $\Delta > 0$ , as the result that there exist roots for the quadratic equation. In addition, based on (42) and (45), it leads to the following constraint for the auxiliary variable  $f(d_{u,k}^{\text{off}})$ :

$$f(d_{u,k}^{\text{off}}) \leq \frac{\hat{h}_{u,k}}{v_{u,k}}. \quad (58)$$

**Proposition 2:** Given  $t_{u,k}^{\text{Ph-IV}}$  and  $\varsigma_k$ , (55) has a unique root under the constraint (58), which is given by

$$f(d_{u,k}^{\text{off}}) = \frac{\mathcal{B}(\alpha, \beta) - \sqrt{\Delta}}{2\mathcal{A}(\alpha, \beta)}. \quad (59)$$

*Proof:* Since the discriminant of the quadratic equation satisfies  $\Delta > 0$ , there exist two roots, which can be denoted as the smaller one  $f^-(d_{u,k}^{\text{off}}) = \frac{\mathcal{B}(\alpha, \beta) - \sqrt{\Delta}}{2\mathcal{A}(\alpha, \beta)}$  and the greater one  $f^+(d_{u,k}^{\text{off}}) = \frac{\mathcal{B}(\alpha, \beta) + \sqrt{\Delta}}{2\mathcal{A}(\alpha, \beta)}$ . We find that only the smaller root  $f^-(d_{u,k}^{\text{off}})$  satisfies the constraint (58), due to the fact that  $f^-(d_{u,k}^{\text{off}}) \leq \frac{\hat{h}_{u,k}}{v_{u,k}} < \frac{\mathcal{B}(\alpha, \beta)}{2\mathcal{A}(\alpha, \beta)} < f^+(d_{u,k}^{\text{off}})$ . This completes our proof. ■

According to (53) and (59), we derive the unique value of  $d_{u,k}^{\text{off}}$  which satisfies the solution of (52) as follows:

$$d_{u,k}^{\text{off}} = W_{u,k} t_{u,k}^{\text{Ph-IV}} (1 - \varsigma_k) \log_2 f(d_{u,k}^{\text{off}}). \quad (60)$$

We find that value of  $d_{u,k}^{\text{off}}$  is associated with the values of  $\alpha$  and  $\beta$ , with the result that we have the following proposition to exploit the solution  $d_{u,k}^{\text{off}}$  of Problem (ECM-Ê-Sub).

**Proposition 3:** Given  $t_{u,k}^{\text{Ph-IV}}$  and  $\varsigma_k$ , there exist three cases for the solution  $d_{u,k}^{\text{off}}$  of Problem (ECM-Ê-Sub) that need to be discussed:

- Case 1: The optimal solution satisfies  $\sum_{k=1}^K d_{u,k}^{\text{off}} = D_u^{\text{tot}} - \frac{\rho_u(T^{\text{max}} - t_{u,k}^{\text{Ph-I}} - t_{u,k}^{\text{Ph-II}})}{v_u}$ , which means that the left side of constraint (47) is strictly binding. It results in that  $\alpha = 0$  holds. In this case, we can find the value  $\beta$  such that  $\sum_{k=1}^K d_{u,k}^{\text{off}} = D_u^{\text{tot}} - \frac{\rho_u(T^{\text{max}} - t_{u,k}^{\text{Ph-I}} - t_{u,k}^{\text{Ph-II}})}{v_u}$ .
- Case 2: The optimal solution satisfies  $\sum_{k=1}^K d_{u,k}^{\text{off}} = D_u^{\text{tot}}$ , which means that the right side of constraint (47) is strictly binding. It results in that  $\beta = 0$  holds. In this case, we can find the value  $\alpha$  such that  $\sum_{k=1}^K d_{u,k}^{\text{off}} = D_u^{\text{tot}}$ .
- Case 3: The optimal solution satisfies  $D_u^{\text{tot}} - \frac{\rho_u(T^{\text{max}} - t_{u,k}^{\text{Ph-I}} - t_{u,k}^{\text{Ph-II}})}{v_u} \leq \sum_{k=1}^K d_{u,k}^{\text{off}} \leq D_u^{\text{tot}}$ , which means that constraint (47) is strictly binding, such that both  $\alpha = 0$  and  $\beta = 0$  hold.

Recall that Proposition 2 provides the solution  $d_{u,k}^{\text{off}}$  of Problem (ECM-Ê-Sub), which depends on the dual variables  $\alpha$  and  $\beta$ . We then exploit the features of solution  $d_{u,k}^{\text{off}}$  in terms of the impact of  $\alpha$  and  $\beta$  for solving Problem (ECM-Ê-Sub), which are illustrated in the following proposition.

**Proposition 4:** Given  $t_{u,k}^{\text{Ph-IV}}$  and  $\varsigma_k$ , the value of  $d_{u,k}^{\text{off}}$  increases with the increasing of  $\alpha$ , but decreases with the increasing of  $\beta$ .

*Proof:* It can be seen from (60) that the value of  $d_{u,k}^{\text{off}}$  increases with  $f(d_{u,k}^{\text{off}})$ . We only need to analyze the monotonic feature of  $f(d_{u,k}^{\text{off}})$  with respect to  $\alpha$  and  $\beta$ .

The first derivative of  $f(d_{u,k}^{\text{off}})$  with respect to  $\alpha$  and  $\beta$  can be separately expressed as

$$\begin{aligned} \frac{\partial f(d_{u,k}^{\text{off}})}{\partial \alpha} &= \frac{N_E(\hat{h}_{u,k} - v_{u,k}) \ln 2}{\mathcal{F}(\alpha, \beta)} \\ &\times \frac{\mathcal{B}(\alpha, \beta) - \sqrt{\mathcal{B}^2(\alpha, \beta) - 4\mathcal{A}(\alpha, \beta)\mathcal{C}(\alpha, \beta)}}{2\mathcal{A}(\alpha, \beta)\sqrt{\mathcal{B}^2(\alpha, \beta) - 4\mathcal{A}(\alpha, \beta)\mathcal{C}(\alpha, \beta)}}. \end{aligned} \quad (61)$$

$$\begin{aligned} \frac{\partial f(d_{u,k}^{\text{off}})}{\partial \beta} &= \frac{N_E(\hat{h}_{u,k} - v_{u,k}) \ln 2}{\mathcal{F}(\alpha, \beta)} \\ &\times \frac{-\mathcal{B}(\alpha, \beta) + \sqrt{\mathcal{B}^2(\alpha, \beta) - 4\mathcal{A}(\alpha, \beta)\mathcal{C}(\alpha, \beta)}}{2\mathcal{A}(\alpha, \beta)\sqrt{\mathcal{B}^2(\alpha, \beta) - 4\mathcal{A}(\alpha, \beta)\mathcal{C}(\alpha, \beta)}}. \end{aligned} \quad (62)$$

We find that the value of  $\frac{\partial f(d_{u,k}^{\text{off}})}{\partial \alpha}$  is higher than zero, the value of  $\frac{\partial f(d_{u,k}^{\text{off}})}{\partial \beta}$  is lower than zero, with the result that  $f(d_{u,k}^{\text{off}})$  increases with  $\alpha$ , but decreases with  $\beta$ . Therefore, the value of

**Algorithm 1:** Subroutine Search Algorithm to Find  $d_{u,k}^{\text{off}}$  and the Corresponding  $\hat{E}_{\text{tot-sub}}^{\min}$ .

---

```

1: Input: Given computation error  $\varepsilon$ , offloading time  $t_{u,k}^{\text{Ph-IV}}$ , secrecy provisioning  $\varsigma_k$ .
2: Case 1 to find  $d_{u,k}^{\text{off}}$  and the corresponding  $\hat{E}_{\text{tot-sub}}^{\min}$ 
3: Initialization: Initialize  $\alpha = 0$ , upper bound  $\beta^{\text{ub}} = \bar{\beta}$ , lower bound  $\beta^{\text{lb}} = \bar{\beta}$ .
4: while  $\beta^{\text{lb}} < \beta^{\text{ub}}$  do
5:   Update bound value of  $\beta^{\text{bv}} = \frac{1}{2}(\beta^{\text{lb}} + \beta^{\text{ub}})$ .
6:   Compute value of  $d_{u,k}^{\text{off}}$  using (60).
7:   if  $\sum_{k \in \mathcal{K}} d_{u,k}^{\text{off}} > D_u^{\text{off}} - \frac{\rho_u(T^{\text{max}} - t_{u,k}^{\text{Ph-I}} - t_{u,k}^{\text{Ph-II}})}{v_u} + \varepsilon$  then
8:     Update the lower bound  $\beta^{\text{lb}} = \beta^{\text{bv}}$ .
9:   else
10:    if  $\sum_{k \in \mathcal{K}} d_{u,k}^{\text{off}} < D_u^{\text{off}} - \frac{\rho_u(T^{\text{max}} - t_{u,k}^{\text{Ph-I}} - t_{u,k}^{\text{Ph-II}})}{v_u} - \varepsilon$  then
11:      Update the upper bound  $\beta^{\text{ub}} = \beta^{\text{bv}}$ .
12:    else
13:      Set  $d_{u,k}^{\text{off}*} = d_{u,k}^{\text{off}}$ .
14:      Compute value of  $\hat{E}_{\text{tot-sub}}^{\min}$  using  $d_{u,k}^{\text{off}*}$ .
15:    end if
16:  end if
17: end while
18: Output:  $d_{u,k}^{\text{off}*}$  and the corresponding  $\hat{E}_{\text{tot-sub}}^{\min}$ .
19: Case 2 to find  $d_{u,k}^{\text{off}}$  and the corresponding  $\hat{E}_{\text{tot-sub}}^{\min}$ 
20: Initialization: Initialize  $\beta = 0$ , upper bound  $\alpha^{\text{ub}} = \bar{\alpha}$ , lower bound  $\alpha^{\text{lb}} = \underline{\alpha}$ .
21: while  $\alpha^{\text{lb}} < \alpha^{\text{ub}}$  do
22:   Update bound value of  $\alpha^{\text{bv}} = \frac{1}{2}(\alpha^{\text{lb}} + \alpha^{\text{ub}})$ .
23:   Compute value of  $d_{u,k}^{\text{off}}$  using (60).
24:   if  $\sum_{k \in \mathcal{K}} d_{u,k}^{\text{off}} > D_u^{\text{off}} + \varepsilon$  then
25:     Update the upper bound  $\alpha^{\text{ub}} = \alpha^{\text{bv}}$ .
26:   else
27:     if  $\sum_{k \in \mathcal{K}} d_{u,k}^{\text{off}} < D_u^{\text{off}} - \varepsilon$  then
28:       Update the lower bound  $\alpha^{\text{lb}} = \alpha^{\text{bv}}$ .
29:     else
30:       Set  $d_{u,k}^{\text{off}*} = d_{u,k}^{\text{off}}$ .
31:       Compute value of  $\hat{E}_{\text{tot-sub}}^{\min}$  using  $d_{u,k}^{\text{off}*}$ .
32:     end if
33:   end if
34: end while
35: Output:  $d_{u,k}^{\text{off}*}$  and the corresponding  $\hat{E}_{\text{tot-sub}}^{\min}$ .
36: Case 3 to find  $d_{u,k}^{\text{off}}$  and the corresponding  $\hat{E}_{\text{tot-sub}}^{\min}$ 
37: Initialization: Initialize  $\alpha = 0$ ,  $\beta = 0$ .
38: Compute value of  $d_{u,k}^{\text{off}}$  using (60).
39: if  $D_u^{\text{tot}} - \frac{\rho_u(T^{\text{max}} - t_{u,k}^{\text{Ph-I}} - t_{u,k}^{\text{Ph-II}})}{v_u} \leq \sum_{k=1}^K d_{u,k}^{\text{off}} \leq D_u^{\text{tot}}$  then
40:   Set  $d_{u,k}^{\text{off}*} = d_{u,k}^{\text{off}}$ .
41:   Compute value of  $\hat{E}_{\text{tot-sub}}^{\min}$  using  $d_{u,k}^{\text{off}*}$ .
42: end if
43: Output:  $d_{u,k}^{\text{off}*}$  and the corresponding  $\hat{E}_{\text{tot-sub}}^{\min}$ .

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$d_{u,k}^{\text{off}}$  increases with the increasing of  $\alpha$ , but decreases with the increasing of  $\beta$ . This completes our proof. ■

The monotonic feature in Proposition 4 enables us to obtain the optimal value of  $d_{u,k}^{\text{off}}$ . The pseudo code of finding the optimal value of  $d_{u,k}^{\text{off}*}$  and the corresponding  $\hat{E}_{\text{tot-sub}}^{\min}$  is presented in Algorithm 1, which is divided into three independent cases. The key steps are explained as follows.

- **Step 3-Step 17:** In Case 1, we adopt the bisection search to obtain the value of  $\beta$  such that  $\sum_{k=1}^K d_{u,k}^{\text{off}} = D_u^{\text{tot}} - \frac{\rho_u(T^{\text{max}} - t_{u,k}^{\text{Ph-I}} - t_{u,k}^{\text{Ph-II}})}{v_u}$ . We first compute the  $\beta^{\text{bv}}$  and  $d_{u,k}^{\text{off}}$ . Then, we update the lower bound as  $\beta^{\text{lb}} = \beta^{\text{bv}}$  under the condition of  $\sum_{k \in \mathcal{K}} d_{u,k}^{\text{off}} > D_u^{\text{off}} + \varepsilon$ . Next, we update the upper bound as  $\beta^{\text{ub}} = \beta^{\text{bv}}$  under the condition of  $\sum_{k \in \mathcal{K}} d_{u,k}^{\text{off}} < D_u^{\text{off}} - \varepsilon$ . Otherwise, we set  $d_{u,k}^{\text{off}*} = d_{u,k}^{\text{off}}$  and compute the value of  $\hat{E}_{\text{tot-sub}}^{\min}$ . Finally, we obtain the optimal value of  $d_{u,k}^{\text{off}*}$  and the minimum  $\hat{E}_{\text{tot-sub}}^{\min}$ .
- **Step 18-Step 33:** Similar to Case 1, we obtain the value of  $\alpha$  such that  $\sum_{k=1}^K d_{u,k}^{\text{off}} = D_u^{\text{tot}}$ . We first compute the bound value of  $\alpha^{\text{bv}}$  and  $d_{u,k}^{\text{off}}$ . Then, we update the upper bound as  $\alpha^{\text{ub}} = \alpha^{\text{bv}}$  under the condition of  $\sum_{k \in \mathcal{K}} d_{u,k}^{\text{off}} > D_u^{\text{off}} + \varepsilon$ . Next, we update the lower bound as  $\alpha^{\text{lb}} = \alpha^{\text{bv}}$  under the condition of  $\sum_{k \in \mathcal{K}} d_{u,k}^{\text{off}} < D_u^{\text{off}} - \varepsilon$ . Otherwise, we set  $d_{u,k}^{\text{off}*} = d_{u,k}^{\text{off}}$  and compute the value of  $\hat{E}_{\text{tot-sub}}^{\min}$ . Finally, we obtain the optimal value of  $d_{u,k}^{\text{off}*}$  and the minimum  $\hat{E}_{\text{tot-sub}}^{\min}$ .
- **Step 34-Step 40:** Case 3 is viable only if constraint (47) holds, with the result that we derive the values of  $d_{u,k}^{\text{off}}$  and the corresponding  $\hat{E}_{\text{tot-sub}}^{\min}$ .

## B. Analysis of Problem (ECM- $\hat{E}$ -Top)

After obtaining the solutions of Problem (ECM- $\hat{E}$ -Sub), we continue to solve the Problem (ECM- $\hat{E}$ -Top) in the following. Since the solutions of Problem (ECM- $\hat{E}$ -Sub) are obtained through distributed search algorithm, it results in that we cannot mathematically express the objective function  $\hat{E}_{\text{tot-top}}^{\min}$  of Problem (ECM- $\hat{E}$ -Top). An important feature in Problem (ECM- $\hat{E}$ -Top) is that both the values of  $t_{u,k}^{\text{Ph-IV}}$  and  $\varsigma_k$  have an interval, i.e.,  $t_{u,k}^{\text{Ph-IV}}$  falls within the given interval  $[0, T^{\text{max}} - t_{u,k}^{\text{Ph-I}} - t_{u,k}^{\text{Ph-II}}]$  based on constraint (48), and  $\varsigma_k$  falls within the given interval  $[0, 1]$ . Moreover, both  $t_{u,k}^{\text{Ph-IV}}$  and  $\varsigma_k$  under the given intervals independent on other parameters. This feature enables us to solve Problem (ECM- $\hat{E}$ -Top) via a two-dimensional line-searching approach. Algorithm 2 presents the two-dimensional line-searching algorithm for solving Problem (ECM- $\hat{E}$ -Top). The key steps are explained as follows.

- **Step 3:** In Problem (ECM- $\hat{E}$ -Top), we first give the lower bound of the uploading time  $T^{\text{lb}}$  and the SOP  $\varsigma^{\text{lb}}$ .
- **Step 4-Step 8:** Under the given values of  $t_{u,k}^{\text{Ph-IV}}$  and  $\hat{\varsigma}_k$ , we invoke Algorithm 1 to compute the current value of  $\hat{E}_{\text{tot-sub}}^{\min}$ . If it satisfies  $\hat{E}_{\text{tot-sub}}^{\min} < \hat{E}_{\text{top}}^{\text{best}}$ , the current best value is updated by  $\hat{E}_{\text{top}}^{\text{best}} = \hat{E}_{\text{tot-sub}}^{\min}$ , and we obtain the current best solution as  $\Phi = \{d_{u,k}^{\text{off}*}, t_{u,k}^{\text{Ph-IV}*}, \varsigma_k^*\}$ .
- **Step 12-Step 14:** We update the values of  $t_{u,k}^{\text{Ph-IV}}$  and  $\hat{\varsigma}_k$  based on the step size  $\ell_1$  and  $\ell_2$  to traverse all numerical cases. Finally, we obtain the optimal value of Problem (ECM- $\hat{E}$ -Sub) as  $\hat{E}_{\text{tot-sub}}^{\min} = \hat{E}_{\text{top}}^{\text{best}}$ , and the corresponding solution of Problem (ECM- $\hat{E}$ -Sub) as  $\Phi = \{d_{u,k}^{\text{off}*}, t_{u,k}^{\text{Ph-IV}*}, \varsigma_k^*\}$ .

Based on Proposition 3, the proposed distributed search algorithm for solving Problem (ECM- $\hat{E}$ -Sub) is guaranteed to converge to the optimal solutions under the given  $t_{u,k}^{\text{Ph-IV}}$  and  $\varsigma_k$ . Therefore, the proposed two-dimensional line-searching

**Algorithm 2:** Two-Dimensional Line-Searching Algorithm for Solving Problem (ECM-Ê-Top) to Find  $d_{u,k}^{\text{off}*}$ ,  $t_{u,k}^{\text{Ph-IV}*}$ ,  $\varsigma_k^*$  and  $E_{\text{tot-sub}}^{\min}$ .

- 1: **Input:** The upper bound of uploading time  $T^{\text{ub}} = T^{\text{max}} - t^{\text{Ph-I}} - t^{\text{Ph-II}}$ , the lower bound of uploading time  $T^{\text{lb}} = 0$ , the upper bound of SOP  $\varsigma^{\text{ub}} = 1$ , the lower bound of SOP  $\varsigma^{\text{lb}} = 0$ , the step size  $\ell_1$  for updating time, and the step size  $\ell_2$  for updating SOP.
- 2: **Initialization:** Set the current best value of energy consumption  $\hat{E}_{\text{top}}^{\text{best}} = \hat{E}_{\text{top}}^{\text{max}}$ , set the current best solution  $\Phi = \{d_{u,k}^{\text{off}}, t_{u,k}^{\text{Ph-IV}}, \varsigma_k\}$  as  $\Phi = \emptyset$ .
- 3:  $\hat{t}_{u,k}^{\text{Ph-IV}} = T^{\text{lb}}$ ,  $\hat{\varsigma}_k = \varsigma^{\text{lb}}$ .
- 4: **while**  $\hat{t}_{u,k}^{\text{Ph-IV}} \leq T^{\text{ub}}$  **do**
- 5:   **while**  $\hat{\varsigma}_k \leq \varsigma^{\text{ub}}$  **do**
- 6:     Invoke Algorithm 1.
- 7:     **if** Problem (ECM-Ê-Sub) is feasible **then**
- 8:       If  $\hat{E}_{\text{tot-sub}}^{\min}$  output by Algorithm 1 satisfies  $\hat{E}_{\text{tot-sub}}^{\min} < \hat{E}_{\text{top}}^{\text{best}}$ , update the current best value  $\hat{E}_{\text{top}}^{\text{best}} = \hat{E}_{\text{tot-sub}}^{\min}$  and the current best solution  $\Phi = \{d_{u,k}^{\text{off}*}, t_{u,k}^{\text{Ph-IV}*}, \varsigma_k^*\}$ .
- 9:     **else**
- 10:       Problem (ECM-Ê-Top) is infeasible.
- 11:     **end if**
- 12:      $\hat{\varsigma}_k \leftarrow \hat{\varsigma}_k + \ell_2$ .
- 13:   **end while**
- 14:    $\hat{t}_{u,k}^{\text{Ph-IV}} \leftarrow \hat{t}_{u,k}^{\text{Ph-IV}} + \ell_1$ .
- 15: **end while**
- 16: **Output:** The optimal value of Problem (ECM-E-Sub)  $E_{\text{tot-sub}}^{\min} = \hat{E}_{\text{top}}^{\text{best}}$ , the corresponding solution of Problem (ECM-E-Sub)  $\Phi = \{d_{u,k}^{\text{off}*}, t_{u,k}^{\text{Ph-IV}*}, \varsigma_k^*\}$ .

algorithm for solving Problem (ECM-Ê-Top) can converge to the optimal solution of Problem (ECM-E-Sub).

### C. Analysis of Problem (ECM-E-Top)

After obtaining the solutions of Problem (ECM-E-Sub), we continue to solve the Problem (ECM-E-Top). The difficulty for solving Problem (ECM-E-Top) is that we cannot mathematically express the objective function of  $E_{\text{tot}}^{\min}$ . An important feature in Problem (ECM-E-Top) is that the value of  $t^{\text{Ph-I}}$  falls within the interval  $[0, t^{\text{max}}]$  based on constraint (38). As a result, this feature allows us to solve Problem (ECM-E-Top) via a line-searching approach. The details for solving Problem (ECM-E-Top) are presented in Algorithm 3. The key steps are explained as follows.

- **Step 3:** In Problem (ECM-E-Top), we first determine the lower bound  $t^{\text{lb}}$  for line-searching.
- **Step 4-Step 10:** Under the given value of  $t^{\text{Ph-I}}$ , Algorithm 2 is invoked to compute the current value of  $E_{\text{tot-top}}^{\min}$ . If it satisfies  $E_{\text{tot-top}}^{\min} < E_{\text{top}}^{\text{best}}$ , the current best value is updated by  $E_{\text{top}}^{\text{best}} = E_{\text{tot-top}}^{\min}$ , and we obtain the current best solution as  $\Theta = \{t^{\text{Ph-I}*}, d_{u,k}^{\text{off}*}, t_{u,k}^{\text{Ph-IV}*}, \varsigma_k^*\}$ .
- **Step 11-Step 13:** The value of  $t^{\text{Ph-I}}$  is updated based on the step size  $\ell_3$  to traverse all numerical cases. We finally obtain the optimal value of Problem (ECM) as  $E_{\text{tot}}^{\min} = E_{\text{top}}^{\text{best}}$ , and the corresponding solution of Problem (ECM) as  $\Theta = \{t^{\text{Ph-I}*}, d_{u,k}^{\text{off}*}, t_{u,k}^{\text{Ph-IV}*}, \varsigma_k^*\}$ .

**Algorithm 3:** Line-Searching Algorithm for Solving Problem (ECM-E-Top) to Find  $t^{\text{Ph-I}*}$ ,  $d_{u,k}^{\text{off}*}$ ,  $t_{u,k}^{\text{Ph-IV}*}$ ,  $\varsigma_k^*$  and  $E_{\text{tot}}^{\min}$ .

- 1: **Input:** The upper bound  $t^{\text{ub}} = t^{\text{max}}$ , the lower bound  $t^{\text{lb}} = 0$ , and the step size  $\ell_3$ .
- 2: **Initialization:** Set the current best value of energy consumption  $E_{\text{top}}^{\text{best}} = E_{\text{top}}^{\text{max}}$ , set the current best solution  $\Theta = \{t^{\text{Ph-I}}, d_{u,k}^{\text{off}}, t_{u,k}^{\text{Ph-IV}}, \varsigma_k\}$  as  $\Theta = \emptyset$ .
- 3:  $\hat{t}^{\text{Ph-I}} = t^{\text{lb}}$ .
- 4: **while**  $\hat{t}^{\text{Ph-I}} \leq t^{\text{ub}}$  **do**
- 5:   Invoke Algorithm 2.
- 6:   **if** Problem (ECM-Ê-Top) is feasible **then**
- 7:     If  $E_{\text{tot-top}}^{\min}$  output by Algorithm 2 satisfies  $E_{\text{tot-top}}^{\min} < E_{\text{top}}^{\text{best}}$ , update the current best value  $E_{\text{top}}^{\text{best}} = E_{\text{tot-top}}^{\min}$  and the current best solution  $\Theta = \{t^{\text{Ph-I}*}, d_{u,k}^{\text{off}*}, t_{u,k}^{\text{Ph-IV}*}, \varsigma_k^*\}$ .
- 8:   **else**
- 9:     Problem (ECM-E-Top) is infeasible.
- 10:   **end if**
- 11:    $\hat{t}^{\text{Ph-I}} \leftarrow \hat{t}^{\text{Ph-I}} + \ell_3$ .
- 12: **end while**
- 13: **Output:** The optimal value of Problem (ECM)  $E_{\text{tot}}^{\min} = E_{\text{top}}^{\text{best}}$ , the corresponding solution of Problem (ECM)  $\Theta = \{t^{\text{Ph-I}*}, d_{u,k}^{\text{off}*}, t_{u,k}^{\text{Ph-IV}*}, \varsigma_k^*\}$ .

### D. Optimality Analysis of the Proposed Algorithms for Solving Problem (ECM)

To address the formulated Problem (ECM), we equivalently decompose it into two subproblems, i.e., Problem (ECM-E-Sub) and Problem (ECM-E-Top). To solve Problem (ECM-E-Sub), we further decompose it into two subproblems, i.e., Problem (ECM-Ê-Sub) and Problem (ECM-Ê-Top). According to Proposition 1, Problem (ECM-Ê-Sub) is proved to be a convex optimization problem under the given values of  $t_{u,k}^{\text{Ph-IV}}$  and  $\varsigma_k$ , we propose subroutine search algorithm (i.e., Algorithm 1) to find  $d_{u,k}^{\text{off}}$  and the corresponding  $\hat{E}_{\text{tot-sub}}^{\min}$ , which is the optimal solution of Problem (ECM-Ê-Sub). For Problem (ECM-Ê-Top), since both the values of  $t_{u,k}^{\text{Ph-IV}}$  and  $\varsigma_k$  have an interval, and are independent on other parameters, we thus propose a two-dimensional line-searching approach (i.e., Algorithm 2) to find the values of  $t_{u,k}^{\text{Ph-IV}}$  and  $\varsigma_k$  until the optimal solution of Problem (ECM-Ê-Sub) is feasible to Problem (ECM-Ê-Top). Therefore, the proposed two-dimensional line-searching approach can find the optimal solution of Problem (ECM-E-Sub). For Problem (ECM-E-Top), since the value of  $t^{\text{Ph-I}}$  falls within an interval, this feature enables us to find a feasible value of  $t^{\text{Ph-I}}$  via a line-searching approach (i.e., Algorithm 3) such that the optimal solution of Problem (ECM-E-Sub) is feasible to Problem (ECM-E-Top). Therefore, the proposed line-searching approach can find minimum of Problem (ECM-E-Top). In summary, the proposed algorithms (i.e., the combination of subroutine search algorithm and line-searching algorithm) can derive the optimal solution of Problem (ECM).

## VI. NUMERICAL RESULTS FOR PROBLEM (ECM)

This section presents the numerical results to evaluate the effectiveness and efficiency of Problem (ECM).

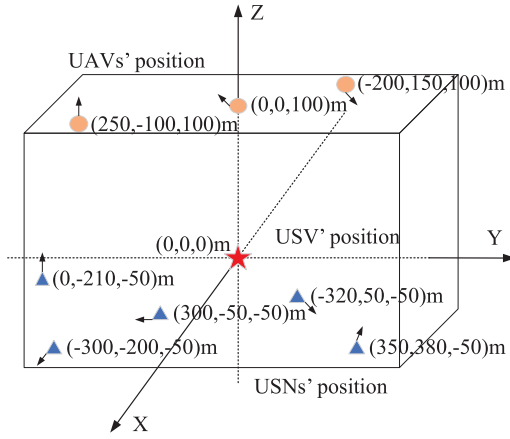


Fig. 2. An illustration of the cuboid space simulation scenario.

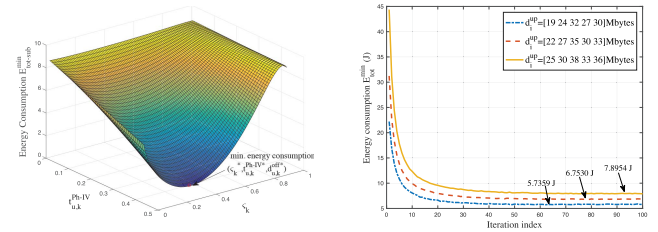
TABLE I  
PARAMETERS USED IN SIMULATION

| Parameters                                                                  | Values          |
|-----------------------------------------------------------------------------|-----------------|
| The fixed height of UAV, $H_k$                                              | 100m            |
| The channel bandwidth of underwater acoustic transmission, $W_i$ [37]       | 1kHz            |
| The channel bandwidth allocated to UAV, $W_{u,k}$                           | 1kHz            |
| The spreading factor, $\kappa$ [36]                                         | 1.5             |
| The power consumption coefficient, $\epsilon_u$                             | $10^{-18}$      |
| The path loss exponent, $\eta$                                              | 1               |
| The channel power gain associated with a reference distance, $\chi$         | $10^{-4}$       |
| The average eavesdropping-path strength, $\mu_{u,E}$                        | $10^{-8}$       |
| The central frequency of acoustic signal, $\theta$                          | 1kHz            |
| The number of CPU cycles for processing one bit of acoustic signal, $\nu_u$ | $10^4$ cycles   |
| The processing capability of USV in CPU cycles per second, $\rho_u$         | $10^6$ cycles/s |
| The channel power gain between USV and eavesdropper, $h_{u,E}$              | $10^{-9}$ dB    |
| The powers of background noise at eavesdropper, $N_E$                       | $10^{-9}$ dBm   |
| The powers of background noise at USV, $N_u$                                | $10^{-9}$ dBm   |
| The offset radius of USV, $C_u$                                             | 5m              |
| The offset radius of UAV, $C_k$                                             | 5m              |
| The maximum computation delay, $T^{\max}$                                   | 4s              |

### A. Simulation Setup

The performance of the proposed scheme is estimated in terms of energy consumption, accuracy and computation time of the proposed algorithms via simulations. We consider a cuboid space simulation scenario with a size of  $0.5 \text{ km} \times 0.5 \text{ km} \times 0.1 \text{ km}$  to model the underwater environment, as shown in Fig. 2. One USV is deployed on upper surface and its position is  $(0,0,0)\text{m}$ . Five USNs are deployed on the seabed and their positions are respectively located at  $(-300,-200,-50)\text{m}$ ,  $(0,-210,-50)\text{m}$ ,  $(300,-50,-50)\text{m}$ ,  $(-320,50,-50)\text{m}$ ,  $(350,380,-50)\text{m}$ . Three UAVs hover in the air at a fixed height above the USV, and their positions are  $(-200,150,100)\text{m}$ ,  $(250,-100,100)\text{m}$ , and  $(0,0,100)\text{m}$ , respectively. Other parameters used in the simulation are summarized in Table I.

To evaluate the efficiency and effectiveness of the proposed algorithms, the following schemes are selected as the comparison benchmarks:

(a) The changes of energy consumption in Problem (ECM-E-Sub), where  $t^{\text{Ph-I}} = 1.1\text{s}$  is exploited.(b) The changes of energy consumption in Problem (ECM-E), where the data volume  $d_i^{\text{up}}$  is exploited.Fig. 3. The convergence of the proposed algorithms for minimizing the energy consumption  $E_{\text{tot}}^{\min}$ .

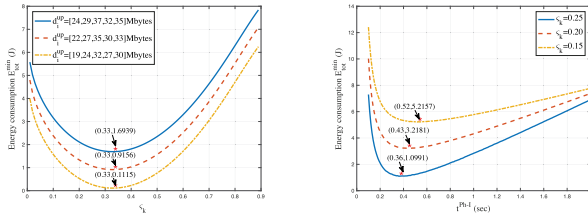
- **FDMA-based Scheme for Underwater Transmission:** USNs upload their collected data to USV via FDMA without any interference. The bandwidth of acoustic channel is uniformly distributed to each USNs.
- **Heuristic Fixed Computation Offloading:** USV can offload its computation workloads to multi-UAVs with the fixed ratio of 25%, 50% and 75%, and uniformly offload to multi-UAVs and USV for local computing.
- **LINGO Global-solver:** The LINGO global-solver adopts the classic algorithm to solve the nonconvexity of the optimization problem, which provides the optimal solution for Problem (ECM) as a benchmark.

### B. Performance Evaluation

Fig. 3 illustrates the effectiveness of the proposed algorithms for minimizing  $E_{\text{tot}}^{\min}$ . Fig. 3(a) exploits the two-dimensional line-searching results of Problem (ECM-E-Sub) by giving the value of  $t^{\text{Ph-I}} = 1.1\text{s}$ . Fig. 3(b) exploits the convergence of Problem (ECM-E) by changing the data volume  $d_i^{\text{up}}$ . It can be seen that the proposed algorithms can minimize the total energy consumption and converge to the minimum value. The higher data volume will lead to a larger energy consumption. The reason is that the processing and offloading transmission power will increase when increasing the data volume.

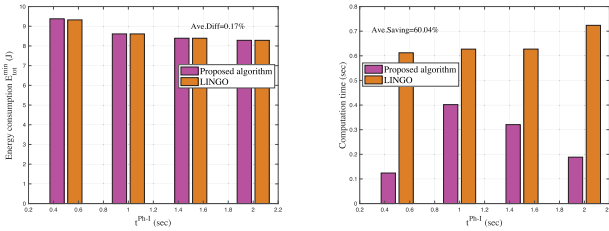
Fig. 4 shows the effectiveness of the proposed algorithms for minimizing the energy consumption  $E_{\text{tot}}^{\min}$ . Fig. 4(a) is the changes of energy consumption  $E_{\text{tot}}^{\min}$  by line-searching the value of  $\zeta_k$ , where the data volume  $d_i^{\text{up}}$  is exploited. Fig. 4(b) is the changes of energy consumption  $E_{\text{tot}}^{\min}$  by line-searching the value of  $t^{\text{Ph-I}}$ , where the data volume  $d_i^{\text{up}}$  and the secrecy-outage probability  $\zeta_k$  are respectively exploited. Combining Figs. 4(a) and 4(b), it can be seen that our algorithms can obtain the minimum value of Problem (ECM).

Fig. 5 verifies the accuracy and computation efficiency of the proposed algorithms in comparison with LINGO. We set the transmission delay in Phase I as  $t^{\text{Ph-I}} = \{0.5, 1.0, 1.0, 1.5, 2.0\}$  sec. The results in Fig. 5(a) show that our proposed algorithms can achieve the solution extremely close to the global optimal solution from LINGO. Meanwhile, the results in Fig. 5(b) further validate the computation efficiency of our proposed algorithms. Thanks to the layered structure,



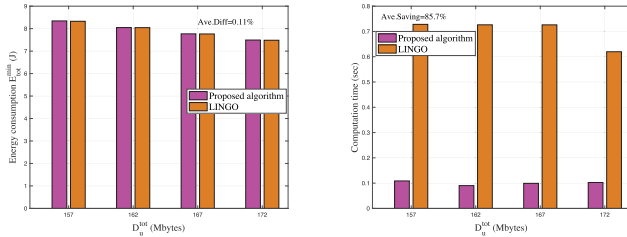
(a) The changes of energy consumption  $E_{\text{tot}}^{\min}$  by line-searching the value of  $s_k$ , where the data volume  $d_k^{\text{dp}}$  is exploited. (b) The changes of energy consumption  $E_{\text{tot}}^{\min}$  by line-searching the value of  $t^{\text{Ph-I}}$ , where the secrecy-outage probability  $s_k$  is exploited.

Fig. 4. The effectiveness of the proposed algorithms for minimizing the energy consumption  $E_{\text{tot}}^{\min}$ , where the values of  $t^{\text{Ph-I}}$ ,  $s_k$  and  $t^{\text{Ph-IV}}$  are exploited.



(a) The accuracy of the proposed algorithms for solving Problem (ECM). (b) The computation efficiency for solving Problem (ECM).

Fig. 5. The accuracy and computation efficiency of the proposed algorithms in comparison with LINGO by changing the transmission delay of Phase I.



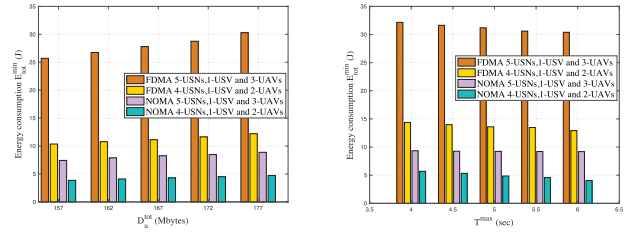
(a) The accuracy of the proposed algorithms for solving Problem (ECM). (b) The computation efficiency for solving Problem (ECM).

Fig. 6. The accuracy and computation efficiency of the proposed algorithms in comparison with LINGO by changing the total amount of transcoding data.

our proposed algorithms can significantly reduce the computational time in comparison with the global-solver of LINGO.

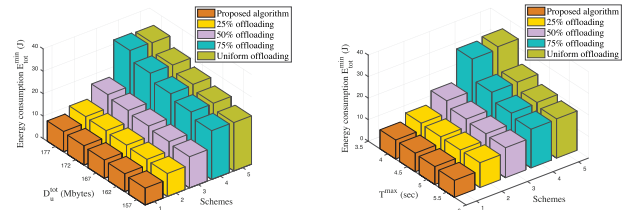
Fig. 6 demonstrates the accuracy and computation efficiency of the proposed algorithms in comparison with LINGO. We set the total amount of transcoding data as  $D_u^{\text{tot}} = \{157, 162, 167, 172\}$  Mbytes. The results in Figs. 6(a) and 6(b) verify the effectiveness of the proposed algorithms. Combining Figs. 5 and 6, thanks to the decomposition approach for Problem (ECM), it is worth pointing that our proposed algorithms can attain the solution extremely close to the global optimal solution, and can significantly reduce the computation time in comparison with the global-solver of LINGO.

Fig. 7 illustrates the performance comparison of the proposed NOMA based data transmission scheme with FDMA based data transmission. The results in Figs. 7(a) and 7(b) demonstrate that our proposed NOMA based data



(a) The comparison of energy consumption  $E_{\text{tot}}^{\min}$  for the proposed NOMA based data transmission and FDMA based data transmission by changing the size of transcoding data. (b) The comparison of energy consumption  $E_{\text{tot}}^{\min}$  for the proposed NOMA based data transmission and FDMA based data transmission by changing the overall delay-limitation.

Fig. 7. The performance comparison of the proposed NOMA based data transmission scheme with FDMA based data transmission.



(a) The comparison of energy consumption  $E_{\text{tot}}^{\min}$  for the proposed offloading scheme, 25% offloading, 50% offloading, 75% offloading, and uniform offloading by changing the transcoding data. (b) The comparison of energy consumption  $E_{\text{tot}}^{\min}$  for the proposed offloading scheme, 25% offloading, 50% offloading, 75% offloading, and uniform offloading by changing the delay-limitation.

Fig. 8. The performance comparison of the proposed offloading scheme with the heuristic fixed offloading scheme.

transmission scheme can obtain a better performance than the FDMA based data transmission scheme. The reason is that the NOMA based data transmission scheme can avoid an orthogonal separation of the bandwidth resource, which improves the transmission throughput in Phase I.

Fig. 8 demonstrates the performance comparison of the proposed offloading scheme with the heuristic fixed offloading scheme. Specifically, “25% offloading”, “50% offloading”, “75% offloading” mean that USV offloads 25%, 50%, 75% workloads to UAVs, respectively. The uniform offloading means that the total workloads are uniformly offloaded to multi-UAVs and USV for local computing. The results in Figs. 8(a) and 8(b) illustrate the advantages of our proposed offloading scheme, which can significantly reduce the total energy consumption. The reason is that the proposed offloading scheme takes the optimal computation offloading into account, with the objective of minimizing the energy consumption.

## VII. CONCLUSION

In this paper, we have investigated UAV-assisted multi-access computation offloading via hybrid NOMA and FDMA in marine networks. In particular, two transmission segments, i.e., the underwater acoustic transmission and the radio frequency transmission, were respectively exploited for computation offloading. In underwater transmission segment, we

considered multiple USNs form the NOMA cluster to upload data to USV via acoustic channels. In radio frequency transmission segment, we considered that USV offloads partial workloads to UAVs for data processing by taking into account the overhearing attack from a malicious eavesdropper in marine environment. To improve the computation offloading efficiency, we have formulated a joint optimization problem of the USN's uploading time, USV's computation offloading, USV's offloading time, and the secrecy provisioning to minimize the total energy consumption of USNs and USV. We have proposed a layered structure to decompose the original problem into a top-problem and a sub-problem, in which efficient algorithms have been proposed to obtain the optimal solutions. Finally, we have conducted extensive numerical results to demonstrate the effectiveness and efficiency of our proposed algorithms compared with benchmark schemes. For the future work, we will investigate multi-USNs' NOMA grouping scenario in underwater environment, in which the coalition formulation of NOMA grouping will be studied.

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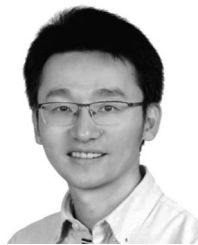
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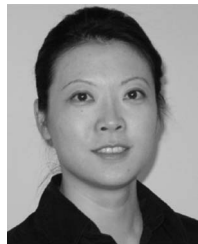


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